

TALKEETNA, AK, USA

WMO: 702510

Lat: 62.320N

Lon: 150.095W

Elev: 350

StdP: 14.51

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26528

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-14.8	-26.9	1.2	-19.4	-21.4	1.7	-9.7	20.0	18.4	18.1	18.1	1.8	50	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	78.2	60.1	74.3	58.7	70.9	57.1	62.3	74.2	60.4	70.4	58.9	67.7	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.7	72.0	64.3	56.2	68.2	63.0	54.8	64.7	61.6	28.0	73.6	26.8	70.5	25.7	67.5	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.3	11.6	DB	-26.4	84.0	8.2	5.0	-32.3	87.6	-37.1	90.6	-41.7	93.4	-47.7	97.1	(4)
(5)			WB	-25.7	64.2	7.0	2.1	-30.8	65.7	-34.9	67.0	-38.8	68.1	-43.9	69.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	14.7	19.5	24.4	36.6	47.9	57.1	60.0	56.7	48.1	34.9	20.9	16.2		
	DBStd	18.80	14.77	12.30	9.87	6.78	5.93	5.10	4.48	4.47	5.49	8.37	11.65	12.96		
	HDD50	5748	1096	854	793	403	111	4	0	3	94	470	872	1048		
	HDD65	10418	1561	1274	1258	852	530	243	165	260	506	934	1322	1513		
	CDD50	823	0	0	0	1	46	216	310	212	37	1	0	0		
	CDD65	20	0	0	0	0	0	6	11	3	0	0	0	0		
	CDH74	383	0	0	0	1	25	135	167	55	0	0	0	0		
	CDH80	80	0	0	0	0	3	33	35	9	0	0	0	0		
	WSAvg	4.1	5.1	4.6	4.9	3.9	4.2	4.1	3.5	2.9	2.9	3.3	4.4	4.8		
	PrecAvg	28.6	1.4	1.4	1.3	1.4	1.7	2.4	3.5	4.6	4.3	3.0	1.7	1.9		
	PrecMax	45.0	4.2	7.0	7.2	4.5	3.9	6.5	9.0	8.5	10.2	10.0	7.2	6.7		
(15) Precipitation	PrecMin	15.5	0.0	0.0	0.1	0.1	0.4	0.5	1.1	1.0	0.9	0.3	0.0	0.0		
	PrecStd	6.7	1.0	1.3	1.4	1.2	0.7	1.5	1.9	2.1	2.5	1.6	1.8	1.5		
	DB	41.2	42.6	49.0	62.6	77.6	84.4	84.8	80.4	68.9	56.0	45.5	40.9			
	MCWB	35.0	35.3	38.5	46.3	54.6	60.5	63.0	63.4	53.8	46.3	39.7	35.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	37.7	39.0	44.6	56.3	70.8	77.5	78.3	74.3	64.1	51.8	40.8	37.7			
	2% MCWB	33.2	33.9	35.8	42.9	52.0	59.0	61.4	59.4	52.2	43.7	36.4	33.0			
	5% DB	35.5	37.0	41.3	52.4	65.9	73.1	74.4	70.2	60.8	48.8	38.1	35.2			
	5% MCWB	31.8	32.7	33.9	40.5	49.8	57.1	60.1	58.2	51.4	42.4	34.2	31.5			
	10% DB	33.4	34.9	38.3	49.6	61.9	69.2	70.7	66.5	57.7	45.8	35.6	33.0			
	10% MCWB	30.1	31.0	32.1	39.1	48.0	55.4	58.5	56.6	50.1	40.8	32.5	30.2			
	0.4% WB	36.5	37.4	39.5	47.3	56.0	63.9	65.1	64.3	56.1	49.3	40.6	36.8			
	0.4% MCDB	38.8	40.4	46.5	61.6	74.3	80.3	80.2	76.8	63.7	53.6	44.7	39.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.3	35.3	37.1	43.6	53.3	60.4	62.8	61.6	54.2	45.8	37.3	33.9			
	2% MCDB	36.8	37.9	42.8	53.9	67.8	73.6	75.6	70.4	60.5	49.7	40.0	36.0			
	5% WB	32.3	33.3	35.0	41.5	51.1	58.4	61.0	59.7	52.9	43.6	35.0	32.3			
	5% MCDB	35.2	36.2	39.6	50.4	62.3	69.9	71.8	67.2	58.3	47.3	37.3	34.8			
	10% WB	30.0	31.1	33.1	40.0	49.3	56.5	59.4	58.1	51.6	41.7	32.9	30.0			
	10% MCDB	33.2	34.6	37.2	48.3	60.0	66.7	68.5	64.6	56.2	45.1	35.3	33.0			
	MDBR	14.3	17.1	20.1	19.3	20.4	18.2	15.7	15.8	15.8	14.0	13.3	13.8			
	5% DB	10.2	13.2	21.0	25.2	29.9	27.6	25.4	24.2	22.3	17.2	10.0	11.1			
(36) Mean Daily Temperature Range	MCWBR	7.8	9.9	14.1	14.0	14.1	11.6	10.9	11.7	12.4	10.6	7.0	8.9			
	5% WB	9.9	11.2	16.9	22.1	24.8	23.3	21.7	19.6	16.9	12.9	8.0	10.0			
	MCWBR	7.9	8.8	11.4	12.6	11.9	10.3	9.8	10.2	10.1	8.7	6.1	8.4			
	MCWBR	7.9	8.8	11.4	12.6	11.9	10.3	9.8	10.2	10.1	8.7	6.1	8.4			
(40) Clear-Sky Solar Irradiance	taub	0.215	0.272	0.284	0.336	0.368	0.369	0.373	0.357	0.314	0.287	0.228	0.202			
	taud	2.051	2.191	2.235	2.159	2.289	2.362	2.371	2.442	2.546	2.432	2.027	1.909			
	Ebn at Noon	149	218	262	267	269	271	266	261	255	217	133	80			
	Edh at Noon	11	20	28	37	36	35	33	29	21	16	10	7			
(44) All-Sky Solar Radiation	RadAvg	81	306	856	1452	1722	1701	1486	1165	762	353	121	39			
	RadStd	11	36	80	91	142	111	116	115	99	42	16	4			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	+3.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page